frequency as shown in Figure 10.37(a). Non-sinusoidal oscillators generate complex, non-sinusoidal waveforms like square-wave, triangular-wave and sawtooth-wave as shown in Figure 10.36 (b), (c), (d).

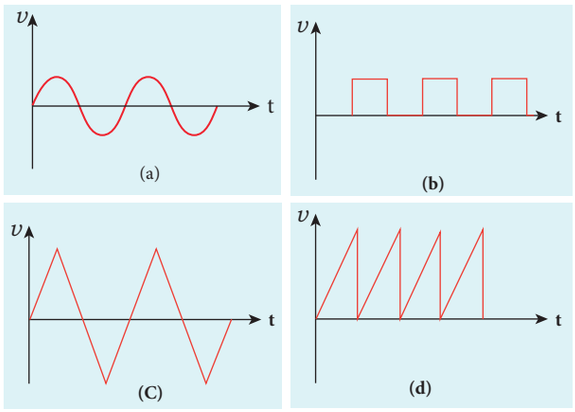


Figure 10.37 (a) Sinusoidal waveform (b) Square waveform (c) Triangular waveform (d)Sawtooth waveform

Sinusoidal oscillations are of two types: **Damped and undamped.** If the amplitude of the electrical oscillations decreases with time due to energy loss, it is called damped oscillations as shown in Figure 10.38(a). On the other hand, the amplitude of the electrical oscillations remains constant with time in undamped oscillations as shown in Figure 10.38(b).

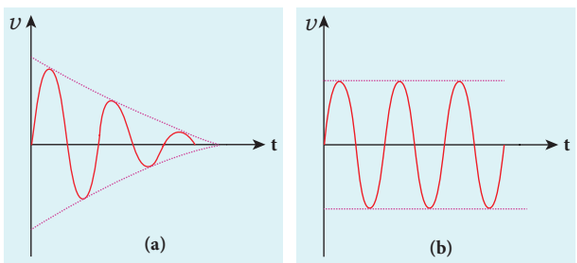


Figure 10.38 (a) Damped oscillations (b) Undamped oscillations

Transistor oscillator

An oscillator circuit consists of three components. They are i) tank circuit ii) amplifier and iii) feedback network. The block diagram is shown in Figure 10.39(a).

i) *Tank circuit*

The *LC* tank circuit consists of an inductance *L* and a capacitor *C* connected in parallel as shown in Figure 10.39(b). Whenever energy is supplied to the tank circuit from a DC source, the energy is stored in inductor and capacitor alternatively. This produces electrical oscillations of definite frequency.

ii) *Amplifier*

This is a single stage amplifier which amplifies the weak signal produced by the tank circuit. The required output is supplied by this amplifier.

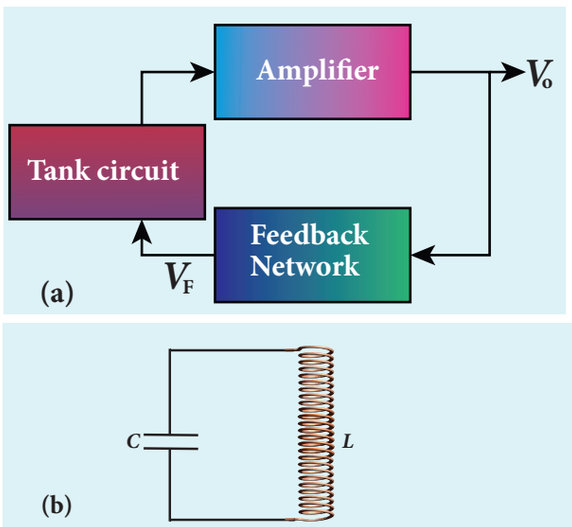


Figure 10.39 (a) Block diagram of an oscillator (b) Tank circuit

iii) *Feedback network*

The circuit used to feed a portion of the output back to the input is called the feedback network. If the portion of the output fed to the input is in phase with the input, then the magnitude of the input signal increases. This process is called positive feedback which is necessary for sustained oscillations.

Working

The tank circuit generates electrical oscillations and acts as the AC input source to the transistor amplifier. Amplifier amplifies

UNIT-10_XII-Physics_Vol-2_New.indd 225 21-07-2020 15:35:36